

1 The join of categories as a pullback over the interval

The *interval category* $\mathbb{I}$ is the category with objects $0, 1$ and a single non-identity morphism $\iota : 0 \rightarrow 1$.

Definition 1. The *join* $\mathbb{C} \star \mathbb{D}$ is the category with objects $\text{ob } \mathbb{C} \sqcup \text{ob } \mathbb{D}$. Its morphisms are those internal to $\mathbb{C}$ or $\mathbb{D}$, plus a unique bridging morphism from any $X \in \text{ob } \mathbb{C}$ to any $Y \in \text{ob } \mathbb{D}$.

The *right cone* $\mathbb{C}^\triangleright := \mathbb{C} \star 1$ freely adjoins a terminal object $\top$ to $\mathbb{C}$. Dually, the *left cone* $\mathbb{D}^\triangleleft := 1 \star \mathbb{D}$ freely adjoins an initial object $\perp$ to $\mathbb{D}$.

We now define two functors to the interval:

- $p : \mathbb{C}^\triangleright \rightarrow \mathbb{I}$ maps $\mathbb{C} \mapsto 0$ and $\top \mapsto 1$. Morphisms internal to $\mathbb{C}$ map to id_0 , and bridging morphisms map to ι .
- $q : \mathbb{D}^\triangleleft \rightarrow \mathbb{I}$ maps $\perp \mapsto 0$ and $\mathbb{D} \mapsto 1$. Morphisms internal to $\mathbb{D}$ map to id_1 , and bridging morphisms map to ι .

Theorem 2. *There is an isomorphism of categories*

$$\mathbb{C} \star \mathbb{D} \cong \mathbb{C}^\triangleright \times_{\mathbb{I}} \mathbb{D}^\triangleleft$$

Proof. We compute the pullback explicitly. Objects in the pullback are pairs (X, Y) where $p(X) = q(Y)$. Morphisms are pairs (f, g) where $p(f) = q(g)$. For objects, we have

- *Over 0:* $p(X) = 0 \implies X \in \text{ob } \mathbb{C}$ and $q(Y) = 0 \implies Y = \perp$. This yields $\{(X, \perp) \mid X \in \text{ob } \mathbb{C}\}$.
- *Over 1:* $p(X) = 1 \implies X = \top$ and $q(Y) = 1 \implies Y \in \text{ob } \mathbb{D}$. This yields $\{(\top, Y) \mid Y \in \text{ob } \mathbb{D}\}$.

Thus, the total object set is precisely $\text{ob } \mathbb{C} \sqcup \text{ob } \mathbb{D}$.

For morphisms, we have

- *Over id_0 :* $f \in \mathbb{C}$ and $g = \text{id}_\perp$, forming a bijection with the morphisms of $\mathbb{C}$.
- *Over id_1 :* $f = \text{id}_\top$ and $g \in \mathbb{D}$, forming a bijection with the morphisms of $\mathbb{D}$.
- *Over ι :* f is a unique bridge $X \rightarrow \top$ and g is a unique bridge $\perp \rightarrow Y$. This pair (f, g) corresponds exactly to the unique bridging morphism $X \rightarrow Y$ in $\mathbb{C} \star \mathbb{D}$.

Since composition in the pullback operates componentwise, it matches the composition in $\mathbb{C} \star \mathbb{D}$. Therefore, the bijection is an isomorphism. $\square$

Note that the isomorphism is natural in both $\mathbb{C}$ and $\mathbb{D}$.

Remark 3. Joyal previously noted the functor $\mathbb{C} \star \mathbb{D} \rightarrow \mathbb{I}$, constructed as the join to the maps $\mathbb{C} \rightarrow 1$ and $\mathbb{D} \rightarrow 1$.

2 The simplicial case

The pullback description does not hold for the classical join of simplicial sets, but it does hold for the categorically equivalent *fat join*.

Definition 4. The *join* $X \star Y$ of simplicial sets is defined by the coend

$$(X \star Y)_n = \coprod_{p+q=n-1} X_p \times Y_q.$$

This extends the categorical join: $N(\mathbb{C} \star \mathbb{D}) \cong N(\mathbb{C}) \star N(\mathbb{D})$. As before, $X^\triangleright := X \star \Delta^0$ and $X^\triangleleft := \Delta^0 \star X$.

The classical join does not preserve colimits in either variable. For instance, $- \star \Delta^0$ fails to preserve coproducts: $(\Delta^0 \sqcup \Delta^0)^\triangleright$ has three vertices (two points sharing a cone point), while $\Delta^1 \sqcup \Delta^1$ has four.

Definition 5. The *fat join* $X \diamond Y$ is defined by the pushout

$$\begin{array}{ccc} (\{0\} \times X \times Y) \sqcup (\{1\} \times X \times Y) & \longrightarrow & \Delta^1 \times X \times Y \\ \downarrow & & \downarrow \\ X \sqcup Y & \longrightarrow & X \diamond Y \end{array}$$

where the left vertical map projects onto X and Y respectively.

Since the pushout defining $X \diamond Y$ is built from functors that each preserve colimits in X (coproduct, product with a fixed object) and pushouts commute with colimits, the fat join $- \diamond Y$ preserves colimits in each variable.

Theorem 6. $X \diamond Y \cong X^\triangleright \times_{\Delta^1} Y^\triangleleft$, where the maps to Δ^1 are as in Section 1.

Proof. An n -simplex of Δ^1 is a map $[n] \rightarrow [1]$, determined by a cut $k \in \{0, \dots, n+1\}$ (sending $\{0, \dots, k-1\} \mapsto 0$ and $\{k, \dots, n\} \mapsto 1$). Over this cut, an n -simplex of $X^\triangleright$ is a $(k-1)$ -simplex of X joined to the cone point, and an n -simplex of $Y^\triangleleft$ is the cone point joined to an $(n-k)$ -simplex of Y . A matching pair over cut k therefore specifies a $(k-1)$ -simplex of X and an $(n-k)$ -simplex of Y , with the cone points contributing no data. Summing over all cuts,

$$(X^\triangleright \times_{\Delta^1} Y^\triangleleft)_n \cong X_n \sqcup \left(\coprod_{p+q=n-1} X_p \times Y_q \right) \sqcup Y_n \cong (X \diamond Y)_n.$$

Face and degeneracy maps agree since they act on the X and Y components independently, shifting the cut accordingly. $\square$

Definition 7. The *comparison map* $c : X \diamond Y \rightarrow X \star Y$ is defined as follows. By the pushout in Definition 5, it suffices to specify maps from each constituent that are compatible on the overlap:

- The inclusions $X \hookrightarrow X \star Y$ and $Y \hookrightarrow X \star Y$ handle the $X \sqcup Y$ component.

- The map $\Delta^1 \times X \times Y \rightarrow X \star Y$ sends a triple (t, σ, τ) with $t : [n] \rightarrow [1]$ (cutting at k), $\sigma \in X_n$ and $\tau \in Y_n$ to the (n) -simplex of $X \star Y$ given by $(d^{>k}\sigma, d^{\leq k}\tau) \in X_{k-1} \times Y_{n-k}$.

On the overlap $\{0\} \times X \times Y$, the cut is at $k = n+1$ (everything maps to 0), so $d^{>k}\sigma = \sigma$ and the Y component is empty, recovering the inclusion of X ; symmetrically for $\{1\} \times X \times Y$.

Proposition 8. *The comparison map c is an isomorphism when X and Y are nerves, and a categorical equivalence in general.*

Proof. Nerves. An n -simplex of $N(\mathbb{C}) \star N(\mathbb{D})$ is a functor $[n] \rightarrow \mathbb{C} \star \mathbb{D}$, determined by a cut k together with a functor $[0, k-1] \rightarrow \mathbb{C}$ and a functor $[k, n] \rightarrow \mathbb{D}$. This is exactly the data of an n -simplex of $N(\mathbb{C})^\triangleright \times_{\Delta^1} N(\mathbb{D})^\triangleleft \cong N(\mathbb{C}) \diamond N(\mathbb{D})$ by Theorem 6, and c sends each such simplex to the corresponding functor. This bijection respects face and degeneracy maps (which restrict or repeat elements of $[n]$, adjusting the cut and component functors), so c is an isomorphism.

General case. Since $\Delta^p = N([p])$, the nerve case gives isomorphisms $c_{\Delta^p, \Delta^q} : \Delta^p \diamond \Delta^q \xrightarrow{\sim} \Delta^p \star \Delta^q$ for all p, q . Every simplicial set is a colimit of representables, and both the fat join (by colimit preservation) and the classical join (by its coend formula) commute with this colimit:

$$X \diamond Y \cong \operatorname{colim}_{\Delta^p \rightarrow X, \Delta^q \rightarrow Y} \Delta^p \diamond \Delta^q, \quad X \star Y \cong \operatorname{colim}_{\Delta^p \rightarrow X, \Delta^q \rightarrow Y} \Delta^p \star \Delta^q.$$

The naturality of c gives a map between these colimits, induced levelwise by the isomorphisms c_{Δ^p, Δ^q} . A colimit of isomorphisms is an isomorphism, so c is an isomorphism. Hence c is in particular a categorical equivalence (see also [3, Proposition 4.3.3.24]). $\square$

Remark 9. In summary, the pullback $X^\triangleright \times_{\Delta^1} Y^\triangleleft$ computes the fat join, not the classical join. These coincide for nerves of categories and agree up to categorical equivalence in general.

References

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